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TEXAS
The University of Texas at Austin

SPECULATIVE DECODING UNDER PRODUCTION SERVING STACKS

Does the fastest draft method stay fastest when engine and load change?
Project-reported range: 2.48× to 0.90× · Llama-3.1-8B · NVIDIA L40 · vLLM + SGLang

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One token commits before the next step can start

MOTIVATION · THE BOTTLENECK

KV caching avoids recomputing the prompt, but it does not remove the token-to-token dependency.

Forward pass Target · 8B	→	**Forward pass** Target · 8B	→	**Forward pass** Target · 8B	→	**Forward pass** Target · 8B	→	**Forward pass** Target · 8B
t1		**t2**		**t3**		**t4**		**tn**

Step i+1 cannot begin until token i is known. One full 8B target step per subsequent token.

Speculation tries to amortize that expensive target-model step across several candidate tokens.

Draft several; verify together

BACKGROUND · HOW IT WORKS

01 · Draft

Propose

A cheaper mechanism predicts several candidate tokens ahead of the confirmed sequence.

→

02 · Target

Verify

The 8B model scores the candidate positions together in one verification step.

→

03 · Commit

Resolve

Accept the matching prefix; reject from the first mismatch and continue with the target prediction.

Correctness. With the exact acceptance procedure, the target model's output distribution is preserved.

Trade-off. The speedup must repay the cost of the drafter and the larger verification step.

Leviathan et al., ICML 2023

The drafter changes; the target does not

METHODS · FOUR MECHANISMS, ONE 8B TARGET

AR baseline

VLLM + SGLANG

No drafter. The 8B target generates one token at a time.

Classical 1B

VLLM + SGLANG

A separate Llama-3.2-1B model drafts sequentially; the 8B target verifies.

Medusa

VLLM ONLY

Additional heads predict future-token candidates in parallel; continuations use tree-based verification.

EAGLE-3

VLLM + SGLANG

A specialized module performs direct token prediction using multi-layer target-model features.

vLLM ablation: fixed draft lengths K = 1, 3, 5.

SGLang adaptive: a distinct acceptance-driven policy; not pooled with the vLLM grid.

Medusa is N/A on SGLang (not listed in 0.5.17 docs). EAGLE-3 checkpoints are engine-specific and were not mixed.

One workload; two independent servers

EXPERIMENTAL SETUP · MEASUREMENT PIPELINE

INPUT

16 technical prompts

Greedy · T=0 · max 256 new tokens

→

SHARED TARGET

Llama-3.1-8B-Instruct

BF16 · ctx 4096 · NVIDIA L40, 48 GB

→

SERVING STACKS

vLLM 0.27.1 + SGLang 0.5.17

One process at a time · c=1 and c=16

PRIMARY MEASURE

Output tokens/s

LATENCY

TTFT + TPOT

RESOURCES

Peak VRAM

DRAFT BEHAVIOR

vLLM accept counters

Known asymmetry: vLLM classical 1B and Medusa ran with --enforce-eager after CUDA-graph capture aborted; SGLang retained graphs. Every speedup is versus the same engine's AR baseline.

Fixed model; real serving-stack differences

COMPARISON DESIGN · WHAT THE EXPERIMENT CAN ISOLATE

Held fixed

Target: Llama-3.1-8B-Instruct weights + tokenizer

Workload: same prompts, decoding, GPU, and loads

Measurement: shared client and metric definitions

Engine-specific

Execution: scheduler, kernels, CUDA graphs, memory

Coverage: which speculative methods each engine exposes

Artifacts: legal EAGLE-3 checkpoint and configuration

This is an end-to-end serving comparison, not a controlled engine-only experiment. Similar rankings support, but do not prove, an algorithmic advantage.

EAGLE-3 leads on both engines

RESULT · CONCURRENCY 1 · PROJECT-REPORTED · NVIDIA L40

vLLM

EAGLE-3 · K=5

2.48×

SGLang

EAGLE-3 · adaptive

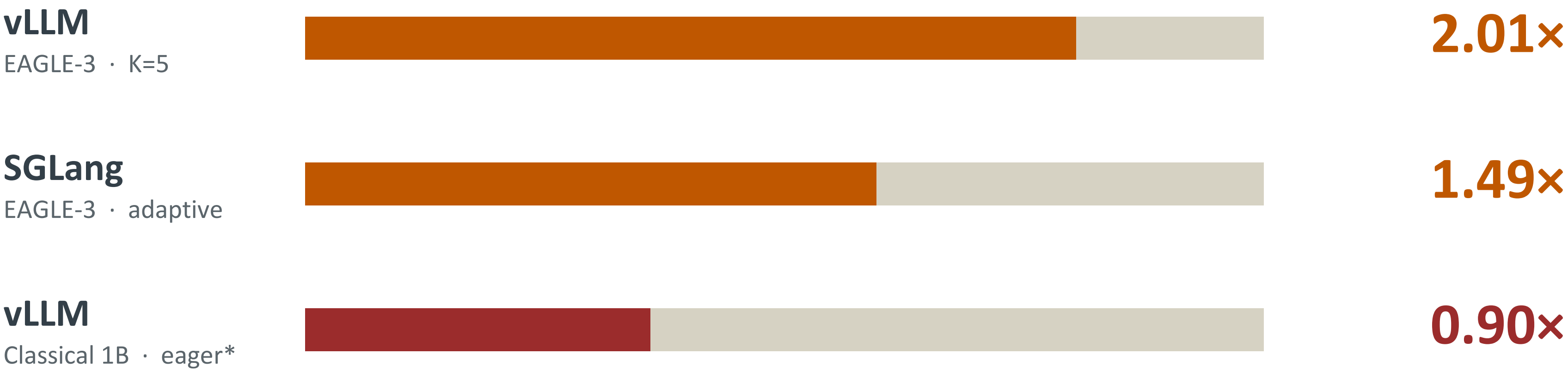
2.36×

Closest cross-engine comparison: EAGLE-3 K=3 reports 2.22× on vLLM and 2.24× on SGLang.

Engine-sensitive case: classical 1B reports 1.05× on eager-mode vLLM versus 1.90× on graph-enabled SGLang.

EAGLE still wins; one configuration regresses

RESULT · CONCURRENCY 16 · PROJECT-REPORTED · NVIDIA L40



The often-cited batch crossover did not occur for EAGLE at concurrency 16 on this L40; it did occur for eager-mode vLLM classical speculation. Throughput: 831.5 tok/s (vLLM K=5) and 628.6 tok/s (SGLang adaptive).

The ranking is clear; the magnitude is stack-dependent

RESULT · FULL MATRIX · PROJECT-REPORTED

tok/s = output throughput · ×AR = same-engine speedup · accept = vLLM mean accepted tokens/draft, c=1 / c=16

vLLM 0.27	c=1 tok/s	×AR	c=16 tok/s	×AR	accept
AR baseline	**40.2**	1.00	413.7	1.00	—
Classical 1B*	**42.3**	1.05	373.6	**0.90**	2.56 / 2.69
Medusa*	**55.9**	1.39	463.7	1.12	0.96 / 0.94
EAGLE-3 K=1	**66.8**	1.66	600.3	1.45	0.76 / 0.75
EAGLE-3 K=3	**89.1**	2.22	741.9	1.79	1.65 / 1.62
EAGLE-3 K=5	**99.5**	**2.48**	**831.5**	**2.01**	2.05 / 2.07

SGLang 0.5	c=1 tok/s	×AR	c=16 tok/s	×AR	accept
AR baseline	**35.1**	1.00	421.0	1.00	—
Classical 1B	**66.9**	1.90	505.5	1.20	—
Medusa	**N/A**	—	N/A	—	not in 0.5.17
EAGLE-3	**78.6**	2.24	589.7	1.40	—
EAGLE adaptive	**83.0**	**2.36**	**628.6**	**1.49**	—

**eager. Acceptance is vLLM-primary; blanks are unavailable, not zero.*

Longer drafts helped through K=5 on this grid

RESULT · EAGLE-3 FIXED-K ABLATION · VLLM

K=1

0.76	1.66×	1.45 ×
accepted	c=1	c=16

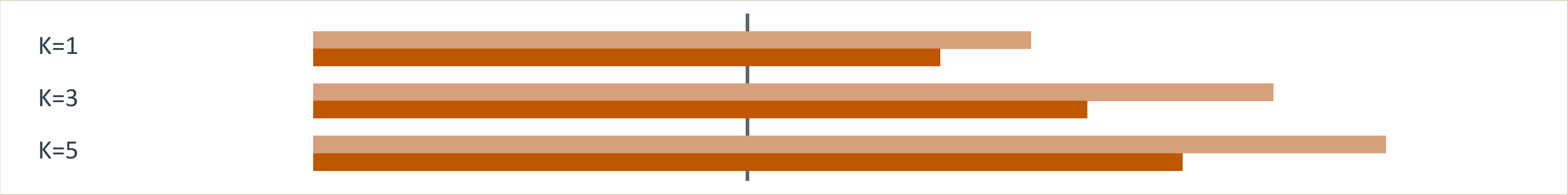
K=3

1.65	2.22×	1.79 ×
accepted	c=1	c=16

K=5

2.05	2.48×	2.01 ×
accepted	c=1	c=16

Speedup versus vLLM AR · light = c=1, solid = c=16 · dashed line = AR baseline



Fixed K values are separate vLLM configurations. SGLang adaptive is not a point on this curve. K>5 was not tested.

Acceptance helps; total cost decides

WHY · QUALIFIED INTERPRETATIONS

speedup = accepted work / (draft cost + verification cost)

DRAFT QUALITY

Medusa accepts 0.96 tokens/draft versus 2.05 for EAGLE K=5; their c=1 speedups are 1.39× and 2.48×.

DRAFT COST

Classical 1B accepts more (2.56) yet reports only 1.05× on eager-mode vLLM. High acceptance alone is insufficient.

SERVING STACK

The same 1B draft reports 1.05× on eager-mode vLLM and 1.90× on graph-enabled SGLang; this gap is confounded.

OFFERED LOAD

EAGLE remains above AR at c=16, but gains shrink as verification work grows with effective batch × K.

Consistent with those mechanisms; the experiment does not separately estimate each component's causal contribution.

The best observed choice depends on engine and load

SERVING IMPLICATIONS · CONDITIONAL ON THIS SETUP

LOAD	VLLM 0.27.1	SGLANG 0.5.17
c = 1	**EAGLE-3, K=5** 99.5 tok/s · 2.48× AR	**EAGLE adaptive** 83.0 tok/s · 2.36× AR
c = 16	**EAGLE-3, K=5** 831.5 tok/s · 2.01× AR	**EAGLE adaptive** 628.6 tok/s · 1.49× AR

Do not generalize the vLLM classical or Medusa rows: both include eager-mode execution. Medusa is N/A on SGLang.

Read the trend, not the third decimal

LIMITATIONS · SCOPE OF THE EVIDENCE

Outside this study: 70B or MoE models · A100/H100 · sampled decoding · production traffic · confidence intervals

What was measured

Llama-3.1-8B-Instruct · NVIDIA L40, 48 GB · BF16 · greedy T=0 · 16 short prompts · vLLM 0.27.1 + SGLang 0.5.17

Statistical limit

One published run per configuration after warmup; no error bars.

Internal confound

vLLM 1B and Medusa used eager mode; SGLang retained graphs.

Asymmetric telemetry

Acceptance analysis is vLLM-primary because equivalent counters were unavailable.

There is no universal speculative-decoding speedup

CONCLUSIONS · ANSWERS TO THE RESEARCH QUESTIONS

2.48× TO **0.90×**

BEST OBSERVED — WORST OBSERVED

Ranking: EAGLE-3 is first on both engines at c=1 and remains first at c=16.

Load: EAGLE gains shrink but remain positive at c=16 on this L40.

Attribution: Acceptance, draft cost, graph mode, checkpoint, and engine implementation all matter.

A useful serving claim names the draft method, engine, hardware, and offered load.

Evidence behind the deck

APPENDIX · REPRODUCIBILITY AND PRIMARY SOURCES

PROJECT ARTIFACTS

Measurements: results/table_latest.json

Shared client: bench/run.py

Server settings: configs/

Coverage: 16/16 requests per published cell

Still needed: raw logs, warmup trace, repeats

PRIMARY SOURCES

Leviathan et al., ICML 2023 — speculative decoding

Cai et al., 2024 — Medusa

Li et al., 2025 — EAGLE-3

vLLM and SGLang speculative-decoding docs

NVIDIA L40 specifications

Benchmark values are project-reported. Architecture and hardware statements are tied to the primary sources above.